

LAIR

B.LAIR_QT

(Booru) Local Archive & Indexing Repository, Qt edition

A desktop booru client. Qt6, C++, no Electron.

VERSION 0.3.0 · MPL 2.0 · re-v0

B.LAIR_QT is native software for browsing boorus with a local gallery. What separates it from other booru clients is that the gallery acts as the booru repository itself: it respects source and tags, and lives in ordinary folders your Linux file manager already understands.

It searches several boorus at once and downloads the originals; tags go into a SQLite index next to the files. Delete the database and the files are still there, still sorted, still openable by anything else on the system.

Boorus that are not on the list are usually one URL away: the app probes the site against six engines and picks the one that fits. Anything else takes a short Python plugin — or, for sites with no API at all, the bundled HTML-scrape bridge.

What it does not do: play video, log into accounts, upload anything, or sandbox plugins.

Quickstart

Arch Linux:

```
makepkg -si
```

Gentoo — drop the ebuild into a local overlay:

```
mkdir -p /var/db/repos/local/media-gfx/blairqt
cp blairqt-0.3.0.ebuild /var/db/repos/local/media-gfx/blairqt/
ebuild /var/db/repos/local/media-gfx/blairqt/blairqt-0.3.0.ebuild manifest
emerge -av media-gfx/blairqt
```

`blairqt-9999.ebuild` builds from git master instead. `USE=python` pulls an interpreter for script plugins; `USE=test` runs the suite during the merge.

Anything else — needs Qt 6 (Widgets, Network, Concurrent, Sql, LinguistTools), CMake 3.21+ and a C++20 compiler:

```
cmake -S . -B build -G Ninja -DCMAKE_BUILD_TYPE=Release
cmake --build build
./build/blairqt
```

On first start a welcome screen asks which sources to load. Nothing explicit is enabled unless you pick it. Type tags, hit Enter, drag the slider if the thumbnails are the wrong size. Right-click a post to download it or to blacklist a tag.

Sources

Six generic parsers — Danbooru, Gelbooru/dapi, Moebooru, e621, Philomena, Szurubooru — cover dozens of sites without a class per site. **Add custom booru** probes a URL and works out which one fits.

Several sources can be merged into one gallery. Paging, infinite scroll and autoscroll are three ways to move through results; picking one disables the controls that stop making sense.

Cloudflare-gated Gelbooru sites accept a cookie and User-Agent you paste in. Sites with no API at all can go through the bundled HTML-scrape bridge — those posts arrive without tags, rating or dimensions, which is the price of scraping HTML.

Every request, status code and response snippet lands in the network log dock. When a site misbehaves, that dock is the first place to look.

Library

Downloads are indexed in SQLite with an FTS5 tag index and written into real nested folders — the same folders Dolphin or Thunar show. Move files between them from inside the app and the tags follow; re-downloading a post you already have is detected and skipped.

Each image also gets a hidden tags sidecar next to it — `image.png` gets `.image.png.txt`, one line of space-separated tags. It follows the file when you move it, and it is what lets another tool (or you, with `grep`) read the tags without touching the database. The metadata is stored as `.txt` as well, because why not — txt is cheap.

Search the library by tag, filter by folder, with or without subfolders. Zoom opens local files instantly, animated GIFs included.

Blacklist

A tag blacklist, global or per source, editable in Settings or straight from a post's context menu. Sites that understand `-tag` get the exclusions pushed into the query; the rest are filtered before they reach the grid.

Matching is exact: `solo` does not hide `solo_focus`. It can apply to the library too, if you want the same rules on what you already downloaded.

Plugins

A new booru engine is one Python file. No C++, no rebuild:

```
def build_search_request(base_url, tags, page, per_page) -> dict:
    """Return {"url": ..., "headers": {...}}."""

def parse_response(body, base_url) -> list[dict]:
    """Return post dicts: id, file_url, preview_url, sample_url,
    post_page_url, tags (list), rating, score, width, height."""

def supports_tag_exclusion() -> bool:      # optional
    """True if the site understands "-tag", so the blacklist can be
    applied in the query instead of only filtering the results."""
```

Add it through **Add custom booru** → *Plugin script*; with Auto-detect on, the script is tried automatically. Tracebacks show up in the dialog log and the network log dock. Plugins need `python3` on `PATH`, and some need their own modules — the Newgrounds example in `examples/` wants `argon2-cffi`.

> **Plugins run with your full user permissions. Only add scripts you trust.**

HTML-scrape bridge

For Gelbooru-engine sites with no `dapi`. Not started automatically:

```
pip install flask cloudscraper beautifulsoup4
python3 tools/html_scrape_bridge.py
```

It listens on `127.0.0.1:8765`. Tick *This site has no API (dapi)* when adding the source.

Appearance and language

Themes are JSON over `QPalette`; 20 curated ones ship with the app and the system default is one click away. Interface languages: Polish, English, German, Danish, Russian, following the system language when it is one of those. Window and dock layout survive a restart.

Safety

The app writes inside its own data folder and folders you explicitly grant, and nowhere else. That is application discipline, not a sandbox — a plugin you add runs as you do. Stored API keys and cookies (`custom_sources.json`) are written owner-only (0600).

Tests

No network, no visible window:

```
ctest --test-dir build --output-on-failure
```

They cover blacklist matching, tag input, response parsing, library paging and the rule that a plugin never blocks the GUI thread.

Data locations

- Settings: `~/.config/blairqt/blairqt.conf`
- Library DB, sources, theme: `~/.local/share/blairqt/blairqt/`
- Thumbnail cache: `~/.cache/blairqt/`

Contributing

Issues and merge requests are welcome. Before sending a patch, run the tests — they take a second and need no network and no display:

```
ctest --test-dir build --output-on-failure
```

Windows port

Welcome, and I will merge it. I do not run Windows, so testing it is yours.

The code does not stand in the way. There are no POSIX calls anywhere, everything goes through Qt, and the binary already looks for its translations and `plugin_runner.py` next to itself — so a portable layout (`blairqt.exe` beside `translations\` and `plugin_runner.py`) works with no changes at all.

What is left to do:

- resolve `python3` to whatever Python is called there (`python.exe`, or the `py -3` launcher),
- hide the tag sidecars — a leading dot hides nothing on Windows, so every image would get a visible `.txt` neighbour,
- bundle the Qt DLLs with `windeployqt` and ship a zip or an installer.

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